

A GPS Based Traffic Light Pre-emption Control System for Emergency Vehicles

Abubakr S. Eltayeb^{#1}, Halla O. Almubarak^{#2}, Tahani Abdalla Attia^{#3}
^{#1,2,3}*Electrical and Electronics Engineering Department, University of Khartoum*
P.O. Box 321, Khartoum, Sudan

¹abubaker@ieee.org

²hosman@uofk.edu

³tahani@uofk.edu

Abstract— Emergency vehicles such as ambulances, police force vehicles and fire fighting vehicles are required to reach their destination as quickly as possible. Any unnecessary delays experienced on them along their way could eventually cause dangerous events; leading to damages in lives or properties. One of the most important delays is the time that is consumed on travelling between the starting point and the destination point of these emergency vehicles; and in intersections with traffic lights, especially when these intersections are congested. In this paper we discuss the design and implementation of an automatic pre-emption traffic control system, based on the global positioning system satellites. It ensures the arrival of emergency vehicles to their destinations on the minimum time possible. The results indicate that the proposed system has an optimum solution for the delay time experienced on the emergency vehicles along their path.

Keywords— Traffic lights, pre-emption, GPS, Delay, Congestion, GSM, Microcontrollers.

I. INTRODUCTION

Reaching the scene of an accident by the emergency team as fast as possible will result in reduction of risks to lives and possessions, where risky situations can be handled properly, fairly minimizing the loss of lives or possessions or at least unwanted complications of injuries. Any delay in the response time of the emergency team will lead to these harmful results. Many factors affect this response time, for example: time for emergency call to be made, time for emergency team to be ready, time for emergency vehicle to reach the location of the accident and time for emergency team to reach the case.

In case of times attached with the emergency team; well training to the team will lead to reducing their response time and reaching the case at the location of the accident. But in case of an emergency vehicle many factors will affect the time duration of travel between its starting position to its desired destination. Such as the distance between the two points of travelling, the speed of the emergency vehicle, the amount of congestion by number of vehicles in the path of the emergency vehicle and the amount of time spent by the emergency vehicle at intersections with red traffic lights on the path of travelling.

This is where several solutions were introduced throughout the years, known as “Pre-emption Systems”, to solve the problem of emergency vehicles path delays. The first pre-emption system was made by manual means, where policemen were distributed in intersections. This solution was

effective in stopping other vehicles outside the road and allowing the emergency vehicle pass through intersections without delays. But sometimes difficulty in recognizing the emergency vehicle by policemen in advance can cause traffic congestion in intersections and this will consume more time. So, traffic laws were introduced after planting sirens, sound and flash light devices, on every emergency vehicle. These laws state that any driver of other vehicle on the way of the emergency vehicle must clear the way for it by all means. Other problems attached to this solution appeared such as accidents that happen when crossing red lights by drivers in order to clear the path for an approaching emergency vehicle, as well as failure in recognizing the sound of the siren, the vehicle’s flashing lights or its approaching direction.

This is where the automatic pre-emption systems were introduced to solve such problems.

II. TECHNICAL BACKGROUND

Currently being used traffic light pre-emption systems can be classified based on its technology of operation as:

1- *Optical systems*: which utilize coded infrared or visual light transmissions from the emergency vehicle to the traffic light in the intersection to provide the path clearance in advance for it. As an example for these systems; OPTICOM, MIRT and TOMAR/SRTOBECOM [3].

2- *Sound based systems*: which they detect the sound from a siren on the emergency vehicle and clear the path accordingly in the intersection. This is done with the use of directional microphones to determine the direction of travel to the intersection. For example to these systems, SONEM 2000 and EPS-II [4].

3- *Radio controlled systems*: In an emergency vehicle, a control button on the emergency unit is pressed to initiate the special timing plan for signals along a selected route. Up to ten intersections could be specified and controlled along the path of travelling. Each pre-determined route had a specific button to activate it. BLISS [5] system is an example to a radio controlled systems.

All of these systems have offered various solutions to the problem from different perspectives. But still other problems are yet to be solved attached to various traffic light pre-emption systems. For example; the narrow range of communication for these systems and decision time based on that range, the repeatedly and unwanted waiting times for

other vehicles' drivers on intersections for the emergency vehicle to pass through and the effect of congested roads, especially during rush-hours, on various pre-emption systems.

III. METHODOLOGY

Our proposed system is giving a unique and optimum solution to the main problem and other problems attached to the different pre-emption systems by:

- a. Suggesting the shortest path with the least obstacles taken from an updated electronic map to the driver of the emergency vehicle, which will result in reducing the emergency vehicles' travelling time to the desired destination.
- b. Clearing the path in advance from other vehicles and pedestrians, especially on traffic lights intersections, which will result in preventing possible congestion, can be made by human estimation or other pre-emption systems for path clearance.
- c. Minimizing the wasted time experienced on other vehicles waiting for the emergency vehicle to pass through, in these intersections.
- d. Preventing accidents that may occur on clearing the path by manual means. Assuring public safety, especially the pedestrians on intersections.

The system utilizes current technology approach, which depends on the international geographical location systems and the global positioning system (GPS)[1], embedded in the vehicle, and a network of traffic lights controllers embedded in the intersections with a reliable communication link method between the traffic lights and the emergency vehicle. Both embedded systems will integrate and develop an instant decision that will help clear the intersection for the emergency vehicle in advance to pass through safely.

The main system components are:

A. Transmitter

On the emergency vehicle side for sending the pre-emption signal that will trigger the system. A Global System for Mobile Communications (GSM) transmitter module is used.

B. Receiver

On the traffic light side for receiving the pre-emption signal from the emergency vehicle. A GSM module is used.

C. Controllers

On both sides for data processing and decision-making. An Atmel family ATmega16 microcontroller is used [8].

D. Road Routing system

With an embedded GPS module is used in the vehicle's side. Providing data such as position, speed, time and heading of a moving object on any place on earth. This GPS technology is available and highly accurate, which our system relies on it, and takes an appropriate decision for pre-emption when the emergency vehicle sends these data to the traffic light controller.

E. Pressure Sensors

Planted on each intersection to count the number of cars available every time. This information is needed for best street calculations for providing a flow of traffic and path clearance in front of an emergency vehicle and for reducing waiting time for other vehicles' drivers.

IV. GPS BASED TRAFFIC LIGHT PRE-EMPTION SYSTEM DESIGN

In the case of emergency; the emergency vehicle (E.V.) driver activates the system of pre-emption when the system allocates the current position of the E.V. using the GPS input data and the driver inputs the location of the destination in the map whether it is a street name, house number, school, station etc. all these are identified to the system by the map as longitude and latitude information. The system calculates the shortest path and starts a communication session with all traffic lights (T.L) along the path while all T.L.s act as listeners to the position of the E.V. This limits the corresponding action of the traffic light just before the E.V. is in the intersection. The flowchart in Fig. 1 explains the main functions of the system.

The approach for designing the system is by dividing it into two major parts:

A. Emergency vehicle side.

B. Traffic light side.

A. Emergency vehicle side

Initially at the vehicle unit, the Visual Basic (VB) program, which was used as a program to open a communication link to the GPS receiver and read the string sent by the National Marine Electronics Association (NMEA 0183) protocol [7]. The extracted information contains the vehicle's longitude, latitude and speed as string of characters in the following format:

xxxx.xxx , yyyy.yyy , sss.sss

Also contains the traffic light's longitude and latitude that are in the E.V. path as string of characters in the following format:

xxxx.xxx , yyyy.yyy

This information is encapsulated as text message and the VB sends the commands using the communication port to initialize the GSM modem to send the message to the desired traffic lights.

The traffic light position is sent only once when first the pre-emption system is activated. While the vehicle position is periodically sent until the pre-emption system is terminated. Finally, VB looks up for the desired SIM card number that represents the traffic light identity to send position information as SMS to the traffic light controller.

Then the program updates the new position by reading a new NMEA string to sends new message containing the current position of the vehicle as shown in Fig. 2.

B. Traffic light side.

This unit contains a microcontroller and GSM modem connected to the traffic light; its function is to receive the position information and response according to it. In order to

do that the traffic light calculates two quantities, estimated Time to Arrive (TTA) and estimated Critical Time (CT). Fig. 3 shows the traffic light controller flow chart.

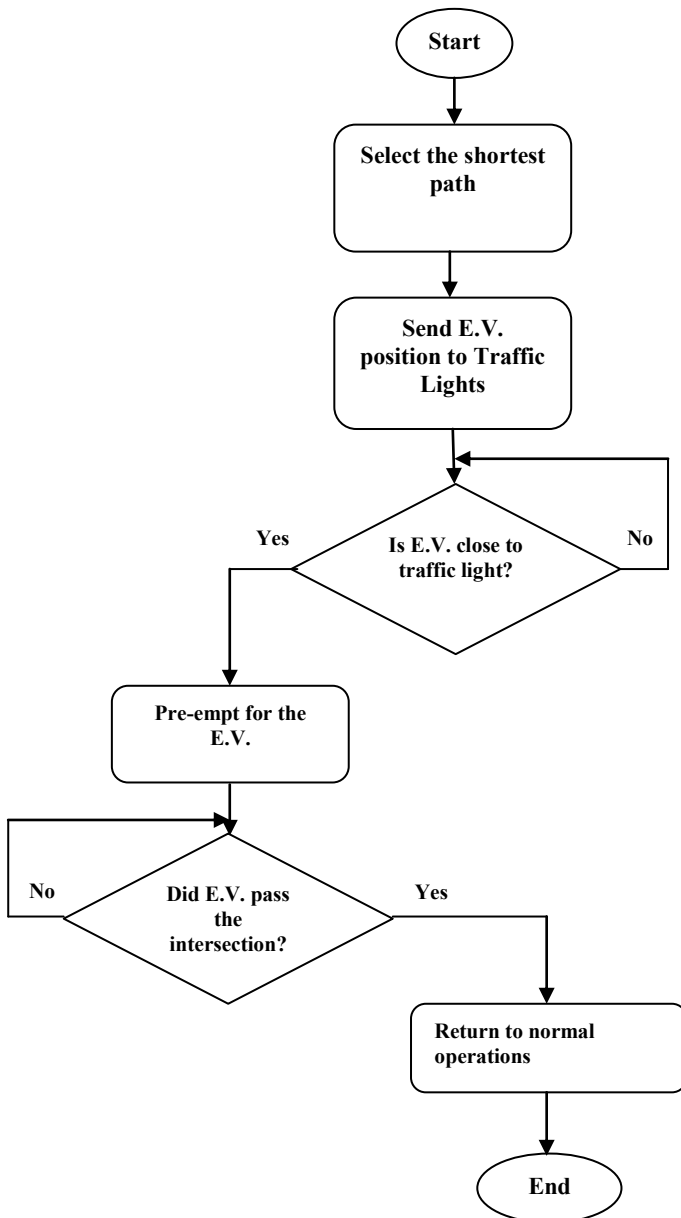


Fig. 1 Main design functions flow chart

- Time to Arrive (TTA):** This quantity of time is calculated by knowing the velocity (V) of the heading vehicle, and applying the velocity law where the distance (D) is the difference between the two positions.

$$D \text{ (meters)} = V \text{ (meters per seconds)} \times \text{Time (seconds)} \quad (1.1)$$
- Critical time (CT):** This quantity of time defines the amount of time necessary for all other vehicles in the desired direction of the intersection to pass the traffic light and clear the path exactly before the E.V. reach the intersection. This amount of time is a statistic value; it

differs from place to place and depends on the structure of the streets, human nature and their responding.



Fig. 2 The GPS Traffic light preemption system user interface

This amount of time is calculated whenever the traffic light is red. The traffic light controller uses a network of sensors planted in the intersection to count the number of vehicles in each intersection. Hence critical time is calculated by knowing the number of vehicles in the intersection and the amount of time it takes a vehicle to change its state from a ready state to a moving state.

Traffic lights in the path are always listeners since the pre-emption system is activated; the microcontroller in the traffic light extracts the SMS received in the GSM modem using the AT commands. The microcontroller calculates the TTA determined by the vehicles position and speed and the traffic light position.

The next step is to calculate the CT by reading the sensors' count for the vehicles existed in the E.V.'s direction at the intersection.

The microcontroller compares these two values, TTA and CT to make one of the following actions:

- The vehicle is far from the intersection, which is the estimated Time to Arrive (TTA) is larger than the estimated Critical Time (CT) then the system turns in listening state keeping the intersection on its normal state (the state of the traffic light before activating the pre-emption system).
- The vehicle is approaching and the TTA is equal to CT, where the system switches to the emergency case by lighting green only to the path of the E.V.
- The E.V. passes the intersection, where the pre-emption system terminates and the traffic light is back to its normal state.

The terminated traffic light is no longer listener and will not receive a position messages from the E.V. because it is now not included in the pre-emption system.

V. SIMULATION AND IMPLEMENTATION

In The simulation program ISIS professional and an Atmega16 microcontroller was programmed to act and response as GSM modem as shown in Fig. 4.

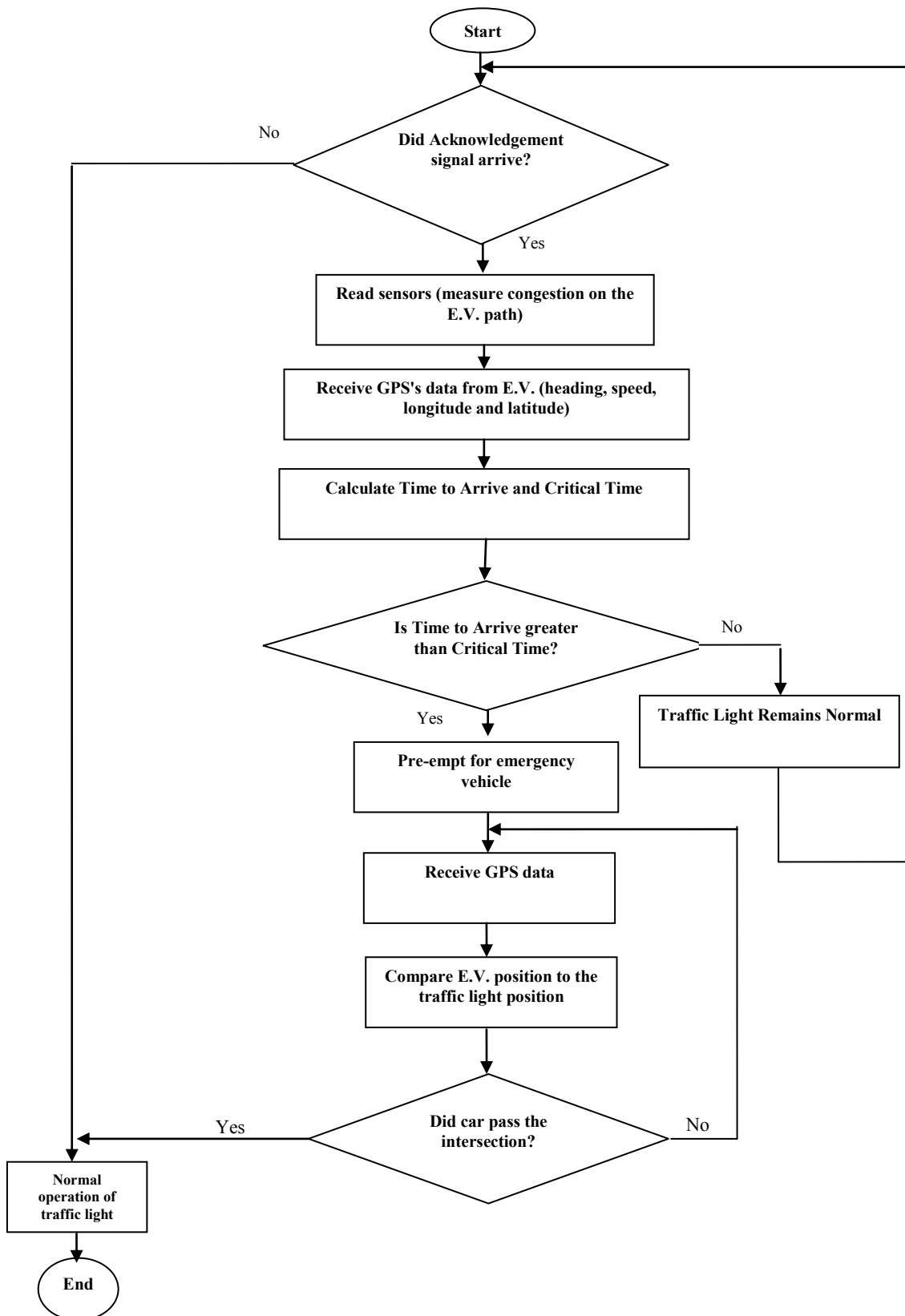


Fig. 3 Traffic light controller flow chart

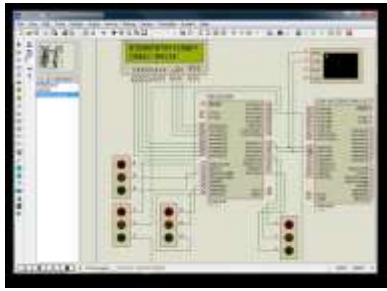


Fig. 4 Traffic Light controller design and simulation

The vehicle unit contains of the following parts:

- A GPS receiver to calculate the E.V. position (Dongle).
- A GSM modem acts as a transmitter to transmit E.V. position
- The circuit of the E.V. unit was constructed as in Fig. 5.



Fig.5 car unit model

The traffic light unit contains of:

- A GSM modem acts as receiver (SIM300S)
- ATmega16 microcontroller.

The circuit of the traffic light unit was constructed as in Fig 6.



Fig. 6 Final circuits operations

VI. RESULTS

The design was tested in several conditions to determine the GPS data of longitude and latitude on field of the experiment and the best time of sending SMS (minimum time of delay upon delivery) and provide a Field experiment of the hardware model. The field of experiment was the street of Othman Digna across Al-Jami'a Street, the west end of the University of Khartoum at the city of Khartoum as shown in Fig. 7.

These locations are shown on Fig. 7 with aid of Google Earth and Google Maps software, distances in meters were measured from the intersection. Table 1 shows results were taken.

In Table 2, three periods of tests were determined, day and date of the experiment and average delivery time in minutes.

Results of the experiment:

- Optimum delivery time was on Saturday 15th of May 2010 at period 1 (8-10 am), average of 0.106 minutes.
- Worst delivery time (maximum delay period) was on Wednesday 5th of May 2010 at period 2(12-2 pm), average of 1.09 minutes. Note: delay time is time the difference between sending time and receiving time.



Fig. 7 Experiment field in Othman Digna Street

TABLE I. GPS READINGS LOCATIONS, DISTANCES FROM TRAFFIC LIGHT AND LONGITUDE-LATITUDE

Map point	Distance from traffic light (meters)	GPS Longitude(E) (degrees)	GPS Latitude(N) (degrees)
A	272	3232.1778	1536.7062
B	254	3232.1787	1536.6957
C	234	3232.1799	1536.6809
D	196	3232.1819	1536.6594
E	172	3232.1840	1536.6444
F	132	3232.1855	1536.6267
G	82	3232.1892	1536.5982
H	36	3232.1917	1536.5744
I	0	3232.1929	1536.5537

VII. DISCUSSION

Field experiment tests on the hardware model on Othman Digna Street across Al-Jami'a Street at the city of Khartoum were successfully accomplished with fair results proved to be compatible with theoretic results to a great extend. The effectiveness of the concept of the proposed solution by the system design was proven by these field experiments. The system behaved as expected in the simulation model. The scenario of the system and its objectives were achieved and tested in real-time environment. The implemented model of the project has encountered delay in response; it can be interpreted from the results, because of the delay of delivery of text messages. This is due to the usage of SMS/GSM network in transmitting and receiving the messages. Various solutions can be made to overcome the delay time using the GSM network. By using a SMS/GPRS technology, an average of 30 messages per minute may be delivered, leading to more accuracy and system prompt responsiveness. Also using another technology for transmitting the data such as radio frequency transmitters and receivers. It also has its advantages, especially in overcoming delay times, and disadvantages such as high cost, but it is gives various solutions and options if the

system is going to be implanted in the future. This also proves the capability of the system design to be developed easily.

TABLE II. AVERAGE DELIVERING TIME FOR MESSAGES

Day	Date	Number of messages sent			Average delivery time (minutes)		
		Period 1 (8-10 am)	Period 2 (12-2 pm)	Period3 (4-6 pm)	Period 1	Period 2	Period 3
Sunday	2/5/2010	12	15	6	0.46	0.61	0.57
Monday	3/5/2010	9	17	10	0.14	0.93	0.37
Tuesday	4/5/2010	7	16	11	0.17	0.9	0.15
Wednesday	5/5/2010	12	22	8	0.49	1.09	0.15
Thursday	6/5/2010	10	18	5	0.2	0.44	0.76
Saturday	8/5/2010	5	10	7	0.133	0.29	0.16
Sunday	9/5/2010	8	20	10	0.2	0.6	0.421
Monday	10/5/2010	10	15	5	0.19	0.98	0.500
Tuesday	11/5/2010	5	15	7	0.19	0.63	0.133
Wednesday	12/5/2010	8	19	8	0.18	0.65	0.721
Thursday	13/5/2010	10	12	7	0.22	1.00	0.16
Saturday	15/5/2010	5	9	5	0.106	0.16	0.2
Sunday	16/5/2010	9	14	7	0.15	0.4	0.18
Monday	17/5/2010	12	14	8	0.15	0.88	0.5
Tuesday	18/5/2010	10	16	10	0.32	0.24	0.86

VIII. CONCLUSIONS

The current paper discusses the design and implementation of a traffic light pre-emption control system that solves an important traffic control problem attached to emergency vehicles. This problem is related to the quick arrival of emergency vehicles to the destination for dealing with dangerous situations that can affect humans' lives and assets. Taking into consideration other unwanted factors such as waiting times to other road users for emergency vehicles to pass through, accidents that may occur when drivers rushes to clear the path for an emergency vehicle as stated in traffic laws. The system is benefiting from GPS (Global Positioning System) technology based on satellites that send accurate information of position, time, and heading with other information for ground receivers. Such technology was adopted into the system because of its accuracy and availability. With such information, emergency vehicles can be described in terms of position as well as the destination point. Also, with aid of software programs attached to the GPS technology, electronic maps were developed to receive such information and display the shortest path to an emergency vehicle to take, saving time of estimation of the path. In accordance to clearing the path; the system adopts controlling the traffic lights of intersection according to time for emergency vehicle to arrive to the intersection, which is sent by a transmitter, and estimated time to clear the

intersection from other vehicles on the way. This estimated time is measured from number of cars queuing. Again, with optimum calculations made by traffic light controller the system responses and achieves the desired objectives. The system with its components has proven its optimal solution to the problem and overcomes other implanted traffic light pre-emption control systems drawbacks. The system is capable of further developments in the future.

REFERENCES

- [1] A.El-Rabbany, Introduction to GPS The Global Positioning System, Artech House, Norwood, Maryland, 2002.
- [2] Gregory T. French, Understanding the GPS: An Introduction to The Global Positioning System what is and how it works, Georesearch Inc., Bethesda, MD, 1996.
- [3] Gene F. Bagnell, An applied research project submitted to the National Fire Academy as part of the Executive Fire Officer Program, USA, 2004.
- [4] (2010) Technology Transfer Center, website. [Online]. Available: <http://www.nttc.edu/ertProgram/evIEWS.asp>
- [5] (2010) ITS America, Emergency Vehicle Signal Pre-emption Systems, website. [Online]. Available: <http://www.itsa.org/committee.nsf/>
- [6] (2010) Technology, Siren Activated Traffic Signals website. [Online]. Available: <http://www.brentwood-tn.org/tech/technology.htm>
- [7] "GPS information & NMEA sentences Data Sheet", The GPS Information website. [Online]. Available: <http://www.gpsinformation.org/dale/nmea.htm>
- [8] "ATmega16 microcontroller Data Sheet", The Compass Lab website. [Online]. Available: <http://www.compass-lab.com/>